Ship Knowledge and Equipment

NTA LEVEL 4 & BRIDGE COURSE

SRT 04104

Lecture Contents

- Introduction to Classification Societies
- International Association of Classification Societies(**IACS**)
- Types of Survey
- Types of surveyors

Introduction to Classification Societies

- What is a classification Society?

Is a non governmental organization that establishes and maintains technical standards for the construction and operation of ships and offshore structures.

The ship owner selects the classification society they want to use and the classification society approves the relevant drawings and inspects the actual construction. Upon completion of construction a certificate is issued

International Association of Classification Societies(IACS)

- Since 1968, the main classification societies have been grouped under **IACS**
- Members of **IACS**
  - ✓ American Bureau of Shipping (ABS)
  - ✓ Bureau Veritas (BV)
  - ✓ Croatia Register of shipping (CRS)
  - ✓ China Classification society (CCS)
  - ✓ DNV GL Group
  - ✓ Indian Registry (IRS)
  - ✓ Korea Register of Shipping (KR)

- ✓ Lloyd's Register (LR)
- ✓ Russian Maritime Register of Shipping(RS)

Types of Survey

- **Initial survey** – A complete inspection of all the items concerning the particular certificate before the ship is put into service to make sure they possess satisfactory conditions and are capable of carry out the service for which the ship is intended.
- **Periodical survey** – Certain certificates require that some or all items related to that certificate need to be inspected at specified intervals within the overall validity pf the certificate. Such an inspection is known as periodic survey.

- **Renewal survey** –Some certificates have a limited validity period. Inspection carried out for the issue of the certificate for a new validity period i.e issuing a new certificate.
- **Intermediate survey** – Refers to inspections that are carried out halfway during the validity period of the certificate. This is done to ensure fitness for service.

- **Annual survey** – Is a general inspection of the items related to the particular certificate carried out annually. This is done to ensure that they have been maintained and remain in a satisfactory condition for the service which the ship is intended. They should be carried out three months before or after each anniversary date of the commissioning or special survey.
- **Additional (Damage) survey** – General or partial Inspection, according to the circumstances, made after a repair resulting from casualty probes or whenever there is an important repair or renewal made.

- **In water or Docking survey**- it is an inspection of the underwater part of the ship and other related items. The inspection is carried out to ensure that the items are in satisfactory condition and fit for the service for which the ship is intended.

Types of surveyors

- What is a marine surveyor?

A Marine Surveyor is a person who conducts inspections, surveys or examinations of marine vessels to assess, monitor and report on their condition and the products on them, as well as inspects damage caused to both vessels and cargo.

General duties of a Classification Society marine surveyor

- Conduct surveys throughout the ship's life (building new ship, annual survey, interim survey, special survey) to ensure standards are maintained.
- Perform inspections required by domestic statutes and international conventions by the International Maritime Organization (IMO).
- Witness tests and operation of emergency and safety machinery and equipment.
- Measure ships for tonnage and survey them for load line assignment.

- Attend court as an expert witness and assist in coroner's inquiries;
- Investigate marine accidents.
- Determine "Fair Market Value, "Damage Repair Costs", and Replacement Value".

- Types of marine surveyors include,
  - ✓ Government surveyor
  - ✓ Cargo surveyor
  - ✓ Classification surveyor
  - ✓ Independent surveyor
  - ✓ Yacht and small craft surveyor

- Government surveyor

A government surveyor performs ship registration surveys, surveys of foreign-going ships and local draft, and generally enforces ship safety standards to ensure marine industrial safety.

The government surveyor belongs to two groups,

i. Flag state surveyor
ii. Port state surveyor

Flag State surveyors report to the government with whom the vessel is registered.

Port State surveyors report to the government into whose territory the vessel has entered.

- Cargo surveyor

A Cargo surveyor is normally appointed by the Cargo Owner, mostly for Bulk, Grain Cargo. Their job is to perform the draft survey to determine the actual cargo loaded on board. They also confirms that the cargo loading is performed according to the law and is within the loadable limits.

- Classification surveyor

A classification surveyor inspects ships to make sure that the ship, its components and machinery are built and maintained according to the standards required for their class.

Classification surveyors often have two roles: one is as a representative of the classification society; and the other as an inspector on behalf of the country with which the vessel is registered (the flag state).

- Independent surveyor

A Independent marine surveyor may be asked to carry out a wide range of tasks, including examining ships' cargoes or onboard conditions such as fuel quality; investigating accidents at sea (e.g., oil spillages or failure of machinery or structures which are not considered to be critical); and preparing accident reports for insurance purposes, and conducting drought surveys to analyse how much cargo has been lost or gained.

- Yacht and small craft surveyor

Yacht and small craft (Y&SC) surveyors specialize in inspecting smaller vessels, typically less than 24 meters, that are most often used for pleasure boating (both power and sail).

Any Questions, Suggestions, Comments, etc.

THE END

